

Class Clitellata

Clitellates

Occurrence c.8,000 spp. worldwide; in soil, freshwater or marine sediments, on marine fishes

Habitat All except dry regions

Clitellates are characterized by the presence of the "clitellum," a largely dorsal area of glandular skin that secretes a cocoon where fertilized eggs are placed. The eggs hatch into miniature versions of the adults—an adaptation to life on land where aquatic-style larvae cannot survive. This class contains some of the world's most familiar invertebrates—the common

earthworm and its relatives. Earthworms belong to the biggest subgroup, the oligochaetes (comprising about 5,000 species), which live mainly in soil, silt, and freshwater. Leeches form a smaller group (about 500 species), and are either predatory or parasitic. Unlike polychaetes, clitellates lack parapodia and head appendages, and are hermaphrodites. They may be up to 9¾ ft (3 m) long.



EARTHWORMS

Found almost worldwide, these familiar invertebrates are often seen when they emerge from their burrows in damp conditions, or are pulled from safety by birds. As Charles Darwin pointed out, their importance in aerating and enriching soil is enormous. *Lumbricus terrestris* (left) can attain a length of 14 in (35 cm).

MEDICINAL LEECH

Most leeches capable of penetrating human skin live in tropical rain forests, but *Hirudo medicinalis* (left) lives in temperate freshwater, and grows to over 4¾ in (12 cm) long. It was once commonly used by the medical profession as a bloodletting agent. When it bites, a leech may inject both anticoagulants and anesthetics.



HORSE LEECH

Leeches that feed on mammals and fishes are ectoparasites whose hosts survive the attacks. The horse leech, *Haemopsis sanguisuga* (above), found throughout Europe, Asia, and North America, can measure 2¼ in (6 cm) long when at rest.

Roundworms

PHYLUM	Nematoda
CLASSES	2
ORDERS	17
FAMILIES	160
SPECIES	About 26,000

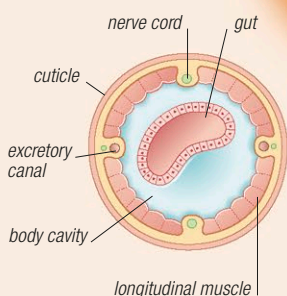
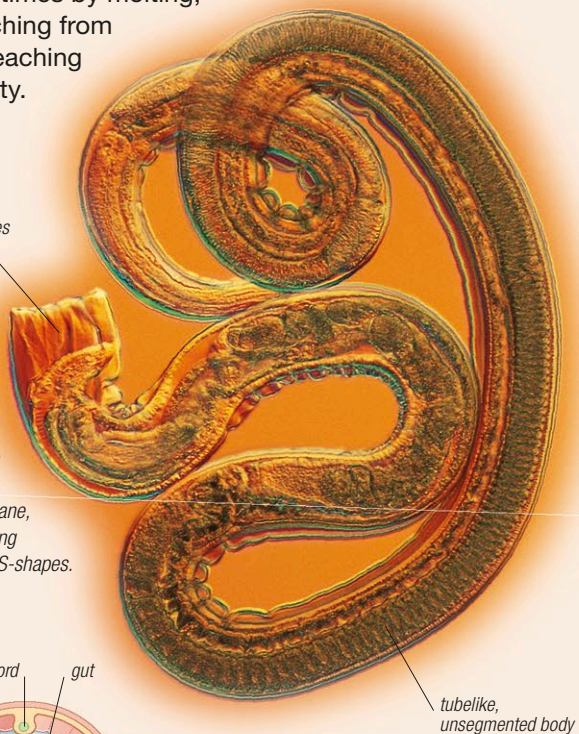
Also known as nematodes, these worms are among the most abundant of all animals. Roundworms are either free living or parasitic. The parasitic forms, which include hookworms, pinworms, and threadworms, live inside almost all types of plants and animals, and some are a significant cause

of disease. Free-living roundworms are also widespread in all aquatic habitats, including small films of water on land. Their bodies are unsegmented, round in cross section, and tapered toward each end. They have a complex cuticle that is replaced 4 times by molting, between hatching from an egg and reaching sexual maturity.

sensory tentacles around mouth

MOVEMENT

Roundworms are unusual in having longitudinal but not circular muscles. They move by thrashing their bodies in a single plane, in the process forming characteristic C- or S-shapes.



BODY SECTION

The large central cavity in a roundworm's body is occupied by fluid-filled cells. The high internal pressure in this cavity, together with a lattice of fibers in the cuticle, prevents the body from shortening when the muscles contract.

Class Secernentea

Secernenteans

Occurrence 14,000 spp. worldwide; in water-filled areas around vegetation and in soil, on plants, in animals



The physical differences between secernenteans and other roundworms, such as adenophoreans (see below), are very technical, relating, for example, to types of sense organs. Secernenteans are mostly parasites, some causing serious diseases. They vary greatly in size, from microscopic to several yards long. The largest known secernentean, up to 30 ft (9 m) in length, lives in the placentas of female sperm whales and is believed to be the largest member of the roundworm phylum. However, the harmful effect of these parasites is not necessarily related to their size: microscopic roundworms, such as pinworms, filarial worms, and hookworms, can multiply rapidly in the host, causing great damage (as in elephantiasis).

WUCHERERIA BANCROFTI

This microscopic roundworm is the cause of elephantiasis in humans, a tropical disease that produces disabling enlargement of soft tissues, particularly in the legs. The worm lives in the lymphatic system, and is spread by mosquitoes.



ASCARIS LUMBRICOIDES

About a sixth of the world's population suffers from ascariasis, an infection by this parasitic roundworm, which can grow up to 16 in (40 cm) long. The infection starts when the eggs are inadvertently taken into the gut with food. The larvae migrate to the lungs, returning to the gut as adults.



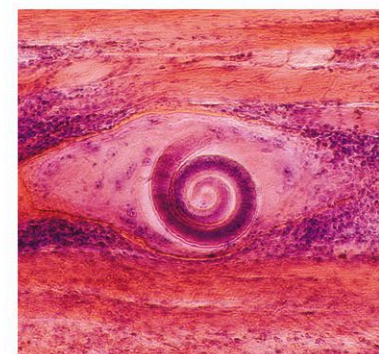
Class Adenophorea

Adenophoreans

Occurrence c.12,000 spp. worldwide; in water-filled areas around vegetation and in soils, on plants, animals

Habitat All except desert regions

Adenophorean roundworms are mainly free living rather than parasitic, and are commonly found in most kinds of habitat throughout the world, particularly in marine sediments. They range in size from microscopic to as long as 3¾ ft (1 m) in exceptional cases. Surveys of seabed mud suggest that vastly more may be awaiting discovery.



TRICHINA WORM

Trichinella spiralis is a small roundworm, ½ in (1 mm) long, which lives mainly in rats and other small mammals. Pigs may pick it up while scavenging and, by eating undercooked pork, each year about 40 million humans do the same.